Safety instructions

User responsibility

- Make sure that the aforementioned **risk assessment** is performed before each use of the robot.
- Make sure that the incorporation of the robot into other applications or ecosystems does not affect the robot's safety aspects

Hardware and/or software modifications

Any modification of the equipment is prohibited.

Niryo relies on the open-source philosophy. Software modifications are permitted. However, the modification of the software elements indicated below is **prohibited**:

- Low-level programming files (stack hardware).
- Configuration of the Raspberry Pi.
- Robot configuration file (speed and acceleration limits, position axes and torque limits).
- Embedded software and configuration of the motor parameters.

Niryo disclaims all liability for any damage caused to and by the robot as a result of these modifications.

Safety symbols

This section specifies all dangers that can arise when doing the work described in the User Manual. Each danger consists of:

- A caption specifying the danger level (DANGER, WARNING, or CAUTION) and the type of danger;
- A brief description of the danger;
- Instruction about how to eliminate danger.

symbol	Designation	Description
	DANGER	Warns that if instructions are not followed, an accident will occur resulting in personal injury.
	WARNING	Warns that if instructions are not followed, an accident will occur which could lead to personal injury or damage to the product.
	CAUTION	Warns of possible damage to the product.
	NOTE	Describes an important fact or condition.
	ADVICE	Describes additional information or an easy way to use the function.
	ELECTROSTATIC DISCHARGE	Warns of a possible electrostatic discharge which could damage the robot.
	ELECTRICAL SHOCK	Warns of an electrical hazard.

symbol	Designation	Description
	CRUSHING OF HANDS	Warns of the risk of crushing hands between two mechanical parts.

General safety information



Before installing, using or programming Ned2, you must read the specifications and the product manual. Install Ned2 and all the electrical equipment in accordance with the specifications and warnings in the Installation, Mechanical interface and Electrical interface chapters. It is important to ensure that any modification to the robot's software will not cause any danger or damage to the user and the product. Children are not permitted to handle the robot.



- Make sure that the robot is properly installed on a firm surface with enough clearance to avoid shocks and collisions. (Refer to part 12.3. Range of motion of the robot).
- Cutting the power or activating the emergency stop button can cause the arm to drop. This drop is cushioned. Do not touch the robot when activating the emergency stop in order to avoid any collision.
- Do not put or install Ned2 in a damp environment or near water.
- Do not install or use Ned2 in hazardous environments (e.g. in a strong magnetic field, hazardous gas, fire or near flammable products) to prevent the hazards that can occur as a result of outdoor conditions.
- Make sure that external modifications to the motor parameters or robot software do not damage the product.
- The robot and motors generate heat when they are running. Do not touch or handle the robot during operation or after prolonged use, the high temperature may cause pain.
- Regularly monitor the motor and arm temperature in Niryo Studio.

- When the Ned2 is activated, keep away from the areas labeled with the pinch warning sticker, to avoid any injury.



-Do not turn off the robot during a sequence or movement, except when absolutely necessary, as this can shorten the lifetime of the robot.
-Preferably switch off the robot using the push button. Avoid using the emergency stop button except in case of danger.
-Failure to follow the connection instructions could damage the tool or robot.

Voltage related risks, robot

A danger of high voltage is associated with the robot in:

- The power supply of the motors (12V , 11.4V, 5V);
- Ned2's grippers are powered by 5V. A risk of short circuit and sparks exists. Be sure to make the installations with the power disconnected.

Safety actions

Fire extinguishing:

> Use a CARBON DIOXIDE (CO_2) extinguisher in the event of a fire in Ned2's system.

Manually stopping or overriding the arm: if needed, Ned2 arm can be stopped manually. This is possible since Ned2 arm is light, the arm force is limited, and the drivetrain power is limited.

To prevent unnecessary damage and wear of Ned2 arm, it is recommended to use the normal stopping functions of Ned2 (by the software or the top button).

> Ned2 is not equipped with brakes since its weight and design do not require a holding brake.

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